EXPERIMENT 10 - LAB RECORD

Performance Optimization in Vehicle Service Request Management System

Efficient request processing and workflow design in a vehicle servicing application

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Application	Vehicle Service Request Management System
Subject	Low Code Application Development using Zoho Creator

1. Aim

To study performance optimization in a Vehicle Service Request Management System by examining how a service request is processed, updated and displayed without repeatedly retrieving unrelated records. The experiment focuses on direct form input, efficient status updates, and avoiding unnecessary loops or duplicate processing.

2. Introduction

Performance optimization in a low-code application involves designing forms, reports and workflows so that each operation processes only the data it needs. Fetching every service request to update one request increases unnecessary work. A more efficient workflow uses the current request's values or a precise criterion to locate the relevant record, validates the required fields, and performs only the required action.

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3. Application used

The practical is represented using a Vehicle Service Request Management System built around a service-request form, a request list/report, and a status board. The workflow receives vehicle and customer details, creates or updates a request, and tracks it through Submitted, Assigned, In Progress and Completed.

Workflow / component	Trigger or purpose	Status in practical
Create_Service_Request	Successful form submission	Core workflow
Validate_Request_Details	Before submission / validation	Core workflow
Update_Service_Status	Technician or admin status update	Core workflow

4. Performance optimization concept

Criteria-based fetching retrieves only the record required for an operation, for example a service request whose Request_ID matches the submitted request ID. Instead of fetching every request and looping through them, the workflow should use the current form's input values when available or query by a unique ID when a separate lookup is necessary.

5. Procedure

1. Open the application — Open the Vehicle Service Request Management System in Zoho Creator edit mode and review the form workflows and related reports.

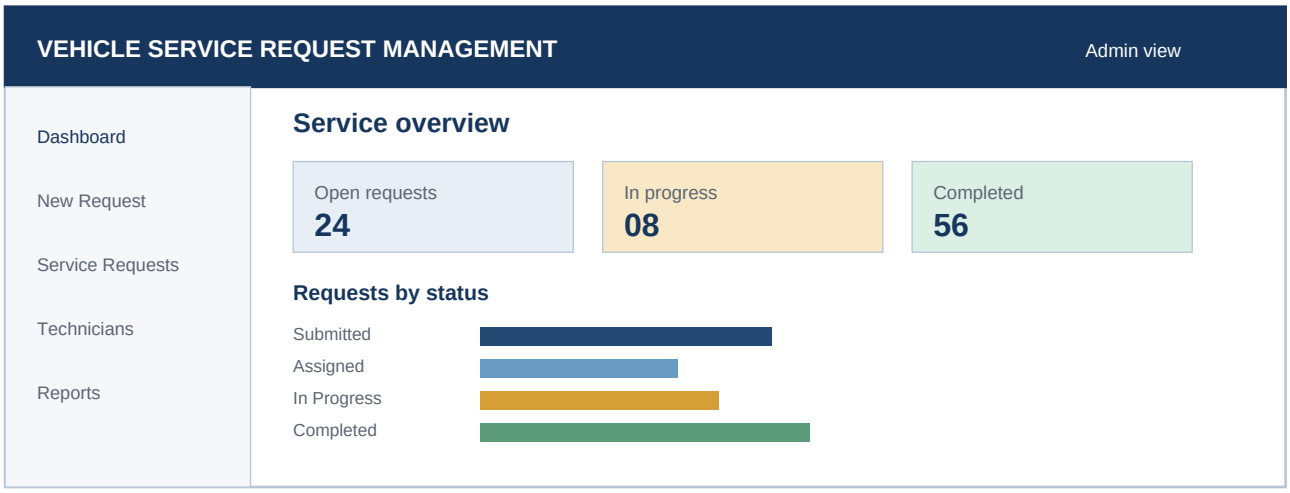

Figure 1: Dashboard showing service-request totals and status summary

2. Review the request form — Inspect the fields used to capture the customer name, vehicle registration, vehicle model, requested service, preferred date and priority. Validate required fields before creating the request.

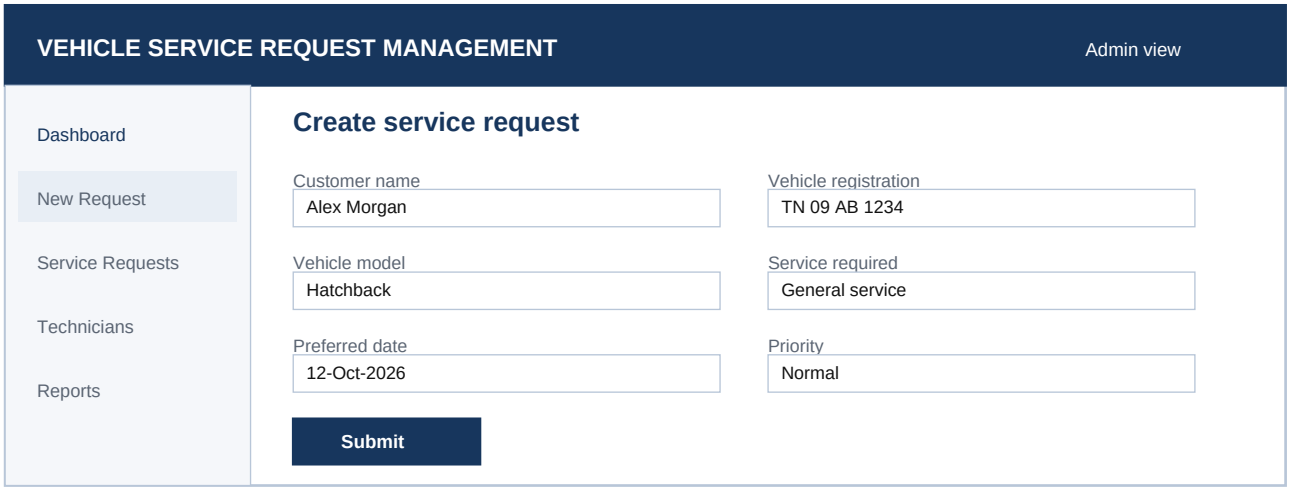

Figure 2: Vehicle service request form with representative sample values

3. Compare request retrieval approaches

The following example is illustrative. It explains the optimization principle rather than claiming that the displayed pseudocode was executed in a live application.

Inefficient approach: retrieve all service requests, loop through the collection, and then identify the one request to update.
Preferred approach: use the current form's input values, or retrieve only the matching request by its unique request ID when a lookup is required.

Approach	Processing pattern
Fetch all records	Retrieve all requests, iterate, locate one request and then process it.
Use criteria / current input	Process the submitted request directly or retrieve only the matching request.

5. Procedure (continued)

4. Review the request-status board — Open the Kanban-style view and confirm that requests are grouped by their current status. A status change should update the relevant request rather than reprocess the full request list.

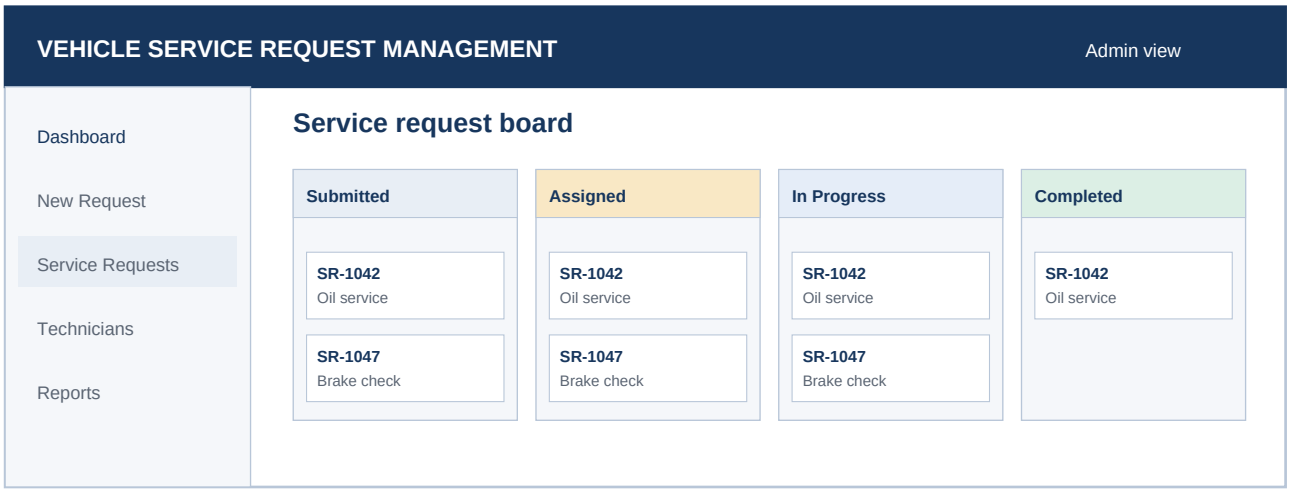

Figure 3: Status board grouping service requests by workflow stage

5. Review the request report — Open the service-request list and verify that each row shows a unique request ID, customer, vehicle, service type and status. Reports should display the required fields and use filters where appropriate.

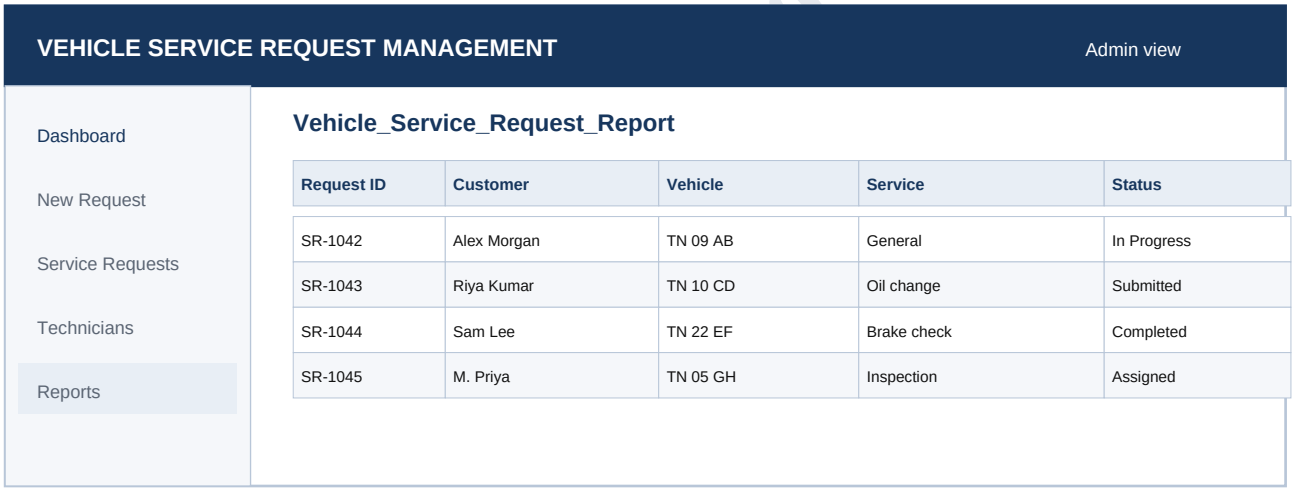

Figure 4: Vehicle_Service_Request_Report with illustrative sample records

6. Test the workflow — Submit a sample request, verify that it appears in the request list, then update its status and confirm that the board and report reflect the same request state.

Note: The application screens above are illustrative mock-ups created for this report, not screenshots captured from a live Zoho Creator application.

6. Analysis of the workflow

Workflow practice	Why it is efficient
Read submitted form values	Uses values already available for the current request and avoids an unnecessary lookup.
Validate required fields once	Prevents incomplete requests from triggering later actions or repeated corrections.
Use a unique request ID	Limits any required lookup to the single relevant service request.
Update only the changed status	Avoids processing unrelated requests when a technician changes one request's stage.
Avoid repeated loops and duplicate actions	Reduces unnecessary processing and lowers the chance of repeated notifications or updates.

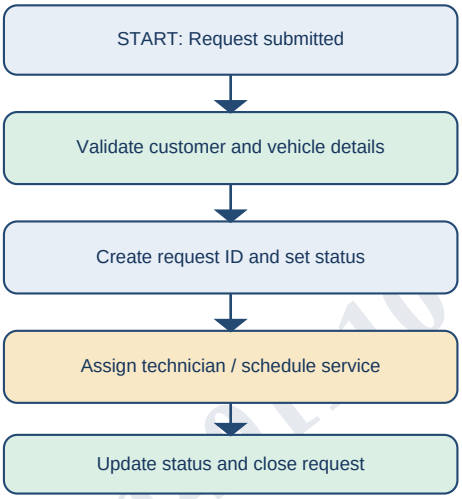

Figure 5: Optimized flow for creating and tracking one service request

The workflow begins when a request is submitted, validates the details, assigns a request identifier, and then supports assignment and status updates. Each step operates on the current request. If a separate record lookup is needed, it should use a specific criterion rather than retrieving the full dataset.

Measurement note: No timed benchmark was performed in this practical. Therefore, no numerical improvement in execution time is claimed. The analysis compares the amount of data and processing required by the two approaches.

7. Optimization techniques

Technique	Explanation	Application in this practical
Criteria-based retrieval	Retrieve only the record needed for the operation.	Use the unique request ID if a separate lookup is required.
Avoid unnecessary loops	Do not scan every request when the current request is already known.	Process the submitted or updated request directly.
Input validation	Check required customer and vehicle details before processing.	Validate request fields before creating the record.
Targeted status updates	Update only the relevant request's status and related fields.	Keep the Kanban board and report consistent with the request record.
Simple workflow design	Avoid duplicate actions and repeated work across workflows.	Keep create, validate and status-update steps focused.

8. Result

• The Vehicle Service Request Management System workflow and its request form, status board and report were examined.
• The optimization approach processes the submitted or updated request directly where possible, rather than fetching and looping through all service requests.
• A unique request identifier can be used to retrieve only the relevant record when a separate lookup is necessary.
• The report and status board provide complementary views for tracking each request through its service lifecycle.
• No benchmark was performed, so no measured execution-time improvement is claimed.

9. Conclusion

This experiment demonstrates that efficient low-code application design depends on limiting each workflow to the data and actions required for the task. In a Vehicle Service Request Management System, direct use of form input, targeted retrieval by request ID, validation of required details and focused status updates help avoid unnecessary processing. These practices can make the application easier to maintain as the number of service requests grows. Any performance benefit should be confirmed with measurements before a numerical improvement is reported.

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